Rapid Communication

Journal of RAMAN SPECTROSCOPY

Received: 8 July 2009 Accepted: 3 November 2009 Published online in Wiley Interscience: 18 January 2010

(www.interscience.wiley.com) DOI 10.1002/jrs.2591

Use of peptide for selective and sensitive detection of an *Anthrax* biomarker via peptide recognition and surface-enhanced Raman scattering

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A short 16-amino acid peptide has been used in place of an antibody to selectively detect the specific *Anthrax* biomarker, protective antigen (PA), using surface-enhanced Raman scattering (SERS). Peptides are more stable than antibodies under various biological conditions and are easily synthesized for a specific target. A peptide that has high affinity to PA was conjugated onto gold nanoparticles along with a Raman reporter and then incubated in various concentrations of PA. Parallel studies in which the peptide sequence was replaced with an antibody were performed to compare the performance of the two methodologies. Both the peptide and antibody functionalized nanoparticles were able to specifically detect PA concentrations down to 6.1 fM. These results demonstrate that these short, robust peptides can be used in the place of traditional antibodies to specifically recognize target biomarkers in the field for the potential diagnosis of disease. **Copyright © 2010 John Wiley & Sons, Ltd.**

Keywords: *Anthrax*; protective antigen; peptide; SERS

Introduction

The detection of specific biomarker proteins has recently been demonstrated to be a reliable method for the early diagnosis of diseases.[1–3] For this and other detection techniques, the selective detection of these target biomarkers must be carried out in complex solution conditions where interfering proteins may compromise positive identification of the analyte at biologically relevant concentrations. Immunoassays which utilize the high selectivity of antibodies are the most popular choice for this type of measurement. Because the concentration of a specific protein biomarker in biological samples is typically sub-nanomolar, highly sensitive detection and tagging methods such as surface-enhanced Raman scattering (SERS) are commonly used to provide the sensitivity required for these types of assays.[4,5] The advantages of SERS arise from the generation of strong electromagnetic fields that are induced by the localized surface plasmon resonance of noble metal nanostructures.[6–8] As a result, the $\sim 10^6$ to 10^{14} fold increase over the normal Raman signal facilitates the detection of target biomarker at ultra-low concentrations.

Although immunoassays have been widely used to selectively recognize a diverse range of protein biomarkers, the biological selectivity of as well as the binding affinity between an antibody–antigen couple can vary significantly upon changes in solution parameters such as pH, ionic strength, and temperature where an antibody could be easily denatured.[9] To combat the problem with antibody stability, short peptides provide a promising and practical alternative for the selective binding and detection of protein biomarkers. First, the peptide structure is more stable in comparison to an antibody in varying pH, ionic strength, and temperature changes. Second, the size of the molecular footprint of the peptide is ~1–2 orders of magnitude smaller than that of an antibody thereby allowing the simultaneous binding of several peptides to different epitopes on a single protein biomarker which potentially leads to improved detection sensitivity. Finally, short peptides are easily synthesized and mass-produced, allowing for their use when high throughput analyses are required.

Herein, we demonstrate that a short 16-amino acid peptide selective to the well-known *Anthrax* biomarker, protective antigen (PA), can be used for the low concentration range detection of target biomarkers using SERS. Previously, short peptides have been utilized to capture *Anthrax* spores in air with high selectivity.[10–12] In this paper, we have further utilized a peptide to selectively recognize the protein biomarker for *Anthrax*, PA, in the liquid phase. The selective recognition of protein biomarkers by short peptides is extremely valuable in the field of bioanalytical detection where the adoptions of traditional immunoassays (antibodies) are the gold standard. The PA-selective peptide sequence, HKHAHNYRLPASGGKK, was chosen using traditional peptide screening in which phage display methods were utilized.

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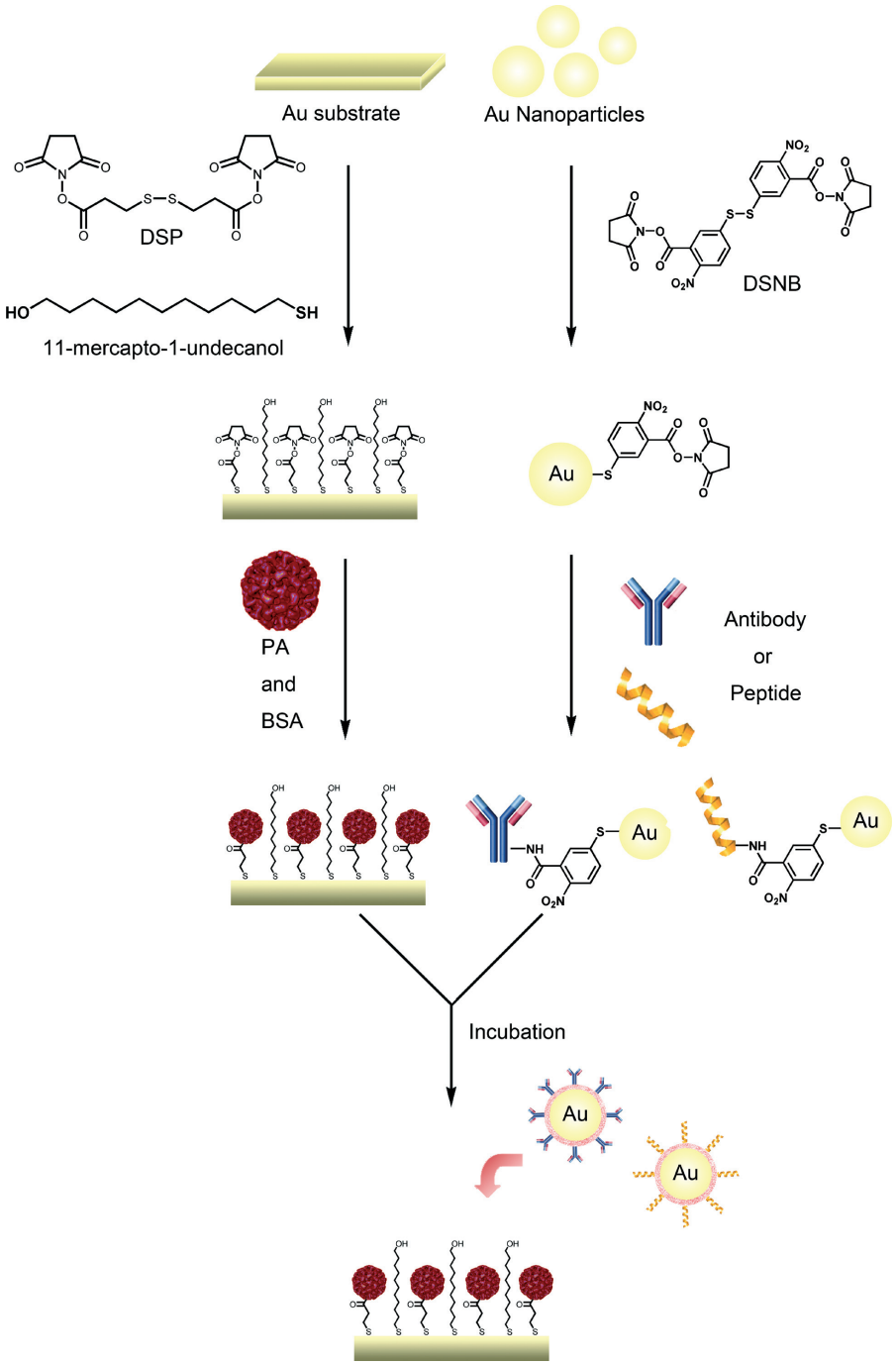

Figure 1. Summary of the conjugation and detection steps in the PA assay. The substrate and Raman-tagged nanoparticle reporters are functionalized in parallel as described in the text. Assay specificity arises from either antibodies or a 16-amino acid peptide that both bind tightly to PA. In the final step of the assay, the two parts are integrated for the quantitative detection of PA using SERS.

The resulting binding affinity between the peptide sequence and PA is 4.06 pM,[13] comparable to the binding affinity of antibody to PA (3.76 pM, a polyclonal IgG antibody) as determined under identical solution conditions.

Experimental

A SERS-based assay for the detection of PA using the selected amino acid sequence along with a traditional antibody assay have been carried out in parallel for comparison purposes. Figure 1 summarizes the overall experimental procedure. First, a self-assembled monolayer of dithiobis-*N*-succinimidyl propionate (DTSP) is formed on the surface of a clean gold substrate via immersion into a 4-mM DTSP solution in DMSO for 4 h. Next, the substrate is rinsed and incubated in 1-mM 11-mercapto-1-undecanol solution in ethanol to passivate empty sites on the metal surface thereby reducing the level of nonspecific binding of proteins or the Raman probe. Eight different concentrations of PA (from 61 nM to 6.14 fM) are applied onto the substrate and allowed

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to bind with the active DTSP binding sites. Finally, the samples are rinsed and the unreacted DTSP molecules are blocked using 2 wt% bovine serum albumin (BSA). A series of phosphate buffered saline solution (PBS) (137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4, 2 mM KH_2PO_4, pH 7.4), 150 mM NaCl solution, nanopure water (18.2 MΩ·cm^{-1}) and tris-buffered saline with Tween-20 (20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 0.05% Tween-20) buffered solution are used in each washing step.

The right-hand side of Fig. 1 outlines the preparation of the SERS probe. First, 200 µl of 2.5 mM 5,5′-dithiobis(succinimidyl-2-nitrobenzoate) (DSNB) solution in acetonitrile is added to 2000 µl of an unconjugated gold nanoparticle (average diameter: 30 nm) suspension and allowed to react for ~4 h. DSNB serves as a Raman reporter molecule which provides a means for quantitative SERS analysis.[14] Next, the DSNB-labeled nanoparticles are centrifuged at 11 000 rpm for 7 min. After centrifugation, the transparent supernatant is removed and the remaining red precipitate (reporter-labeled nanoparticles) is re-dispersed in 2-ml PBS buffered solution. Then, nanoparticles are allowed to separately conjugate to both a polyclonal IgG antibody (rabbit) or the PA-specific peptide sequence. The parallel assays serve as a straightforward comparison of the selectivity and specificity of the two receptor molecules. Equal concentration of both Raman probes are exposed to the previously prepared gold substrates that contained varying concentrations of PA and a control. In all cases, only a gold substrate covered with BSA was used as a control. It should be noted that the size of the peptide in Fig. 1 has been exaggerated for visual presentation as the actual size of the peptide is ~75 times smaller than the antibody.

SERS spectra have been collected using a dispersive Raman microscope equipped with a diode laser (λ_{ex} = 785 nm, power = 44 mW), a thermoelectrically cooled (−40 °C) charge-coupled detection (CCD) detector, and a holographic grating (Kaiser Optical Inc., Ann Arbor, MI, USA). Each PA-captured substrate (either via peptide or antibody) was positioned on a microscope stage and the laser beam was focused using an objective lens (10×/0.25 numerical aperture (NA)) to collect SERS spectra with a resolution of 4 cm^{-1}. For substrate, 20 Raman spectra (exposure time for each spectral collection: 10 s) were automatically collected at a 10-µm interval over a 2000 µm^2 (40 µm × 50 µm) area.

All reported spectra represent an average of 20 spectra collected by mapping. This mapping technique greatly improves the reliability and reproducibility of the measurements.

Results and Discussion

Figure 2 shows the averaged SERS spectra corresponding to eight different concentrations of PA and a control assay using both the peptide (a) and antibody (b) labeled Raman probes. The spectra have been offset for clarity. In both data sets, the primary DSNB bands are located at 1558, 1336, 1061, and 848 cm^{-1}, and are in clear agreement with previously published spectra.[14–16] Both the heights and areas of all of these bands increase with increasing PA concentration for both data sets; while, no significant DSNB bands are observed for both control measurements.

Figure 3 shows the experimental response curves (concentration vs intensity) calculated from the peak height of the 1336 cm^{-1} DSNB band for the peptide (a) and antibody (b) functionalized nanoparticles. The log–log plot insets reveal linear responses for both data sets. The best fit lines in these data sets are represented by the following equations: peptide (a); $y = 0.15x + 0.96$, antibody (b); $y = 0.16x + 0.93$. Both Raman tag approaches exhibit highly linear responses over a wide concentration range (from 61 nM to 6.14 fM) with R^2 values of 0.990 in the peptide case and 0.998 in the antibody case; while, antibody functionalized nanoparticles exhibit slightly better correlation than the peptide functionalized nanoparticles.

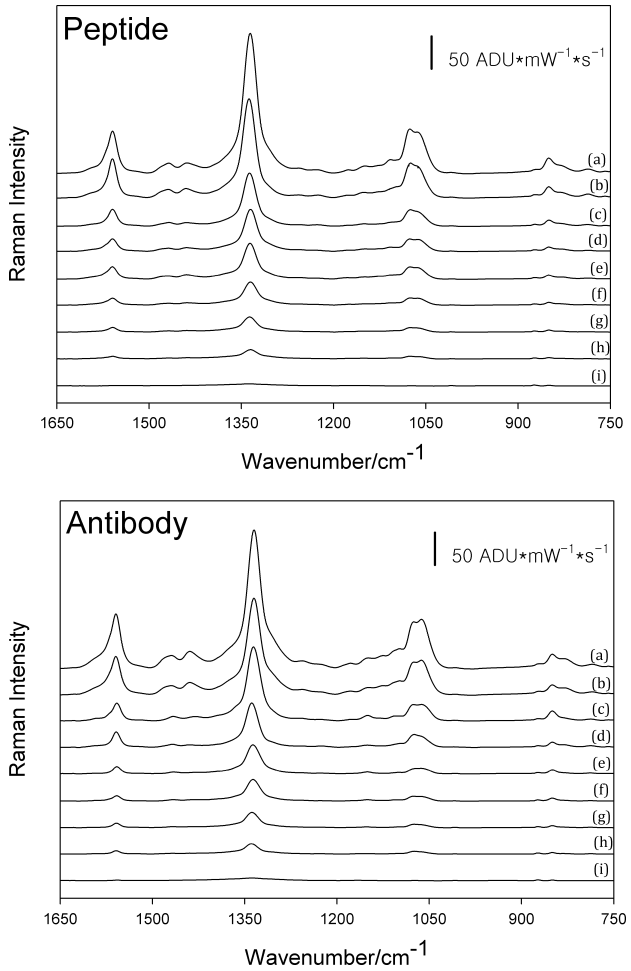

Figure 2. Averaged SERS spectra of the Raman reporter dye, DSNB, in response to varying concentration of the targeted biomolecule PA. The results for peptide (a) and antibody (b) assays are included for their direct comparison. For both (a) and (b), PA concentrations range from (1) 61 nM, (2) 6.1 nM, (3) 610 pM, (4) 61 pM, (5) 6.1 pM, (6) 610 fM, (7) 61 fM, (8) 6.1 fM, and (9) control.

As clearly seen in Fig. 2, the two data sets exhibit similar concentration responses as well as the magnitude of the vibrational intensities are almost similar. Furthermore, the response curves in Fig. 3 exhibit similar response patterns with the similar sensitivities for both cases based on the comparison of the slopes calculated from regression analysis. The intensities corresponding to 6.1 fM in both cases are approximately 4.5 times larger than those of control measurements; therefore, both Raman tag approaches are able to measure PA concentration down to 6.1 fM.

Overall, these results clearly demonstrate the selected 16-amino acid peptide binds strongly to the targeted protein biomarker and exhibits an SERS detection down to 6.1 fM comparable to that of an antibody. While the peptide conjugated system results in slightly inferior concentration correlation in comparison to the parallel antibody assay, the results are still encouraging. We

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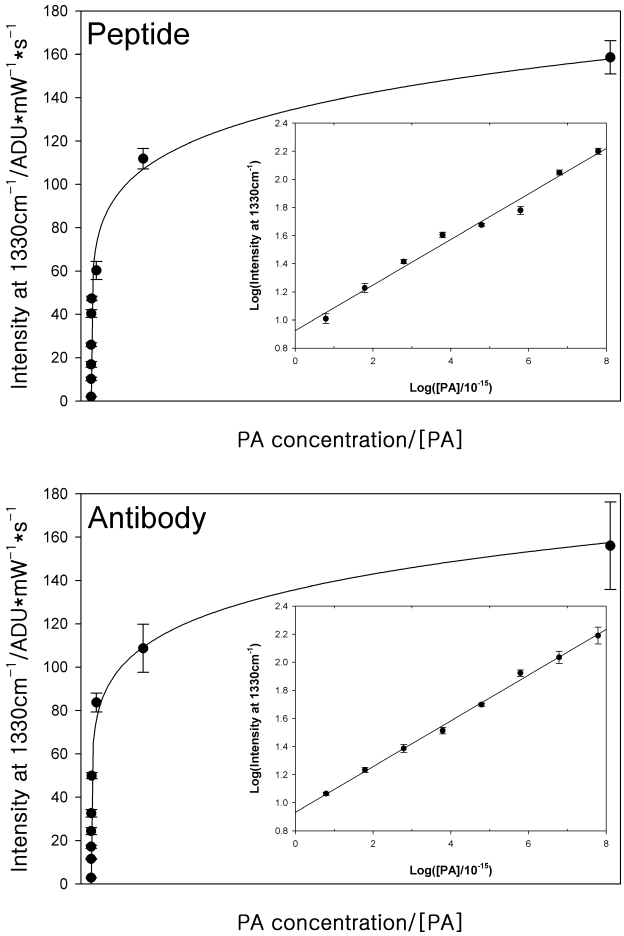

Figure 3. Experimental response curves (concentration vs intensity) obtained from the use of the peptide (a) and antibody (b) functionalized nanoparticles. The corresponding log–log plots are also shown.

hypothesize that the concentration correlation of the peptide could be effectively improved by incorporating multiple selective peptides to bind into more than one epitope on a protein biomarker, thereby improving the detection specificity.

Conclusion

We have successfully demonstrated the feasibility of the conjugation of a 16-amino acid peptide to gold nanoparticles that are labeled with a Raman tag for the detection of a protein biomarker using SERS. Specificity of the detection arises from the surface chemistry (antibody or peptide). Because several different Raman reporters could be used in conjunction with other selective surface chemistries, this type of assay offers the possibility of a multiplexed assay. Furthermore, as additional biomarkers and peptide sequences are identified and/or selected, this analytical approach could be applied to the detection of a wide range of biological and environmental pathogens.

Acknowledgments

This article is dedicated to the memory of the late Kyungtag Ryu, collaborator and colleague of the authors of this article, whose untimely death left his dream of being a great chemist unfulfilled. His passions and endeavors for chemistry will live on through this work.

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